THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $x^2 + 6x + 9$	$(a+b)^2 = a^2 + 2ab + b^2$
2	**(c)** 4 : 1	Convert to the same unit
3	**(d)** 25%	Change/original × 100 = 20/80×100
4	**(b)** $\sqrt{74}$	$h^2 = 5^2 + 7^2 = 74$
5	**(a)** 8:12:15	Make B equal: B = $\text{LCM}(3,4) = 12$; A = 8, C = 15 gives $8 : 12 : 15$
6	**(d)** $x^2 - 4x + 4$	$(a-b)^2 = a^2 - 2ab + b^2$
7	**(b)** 4 : 6 : 5	Make the middle terms equal
8	**(a)** 25%	18/72×100
9	**(c)** 130	Square on hypotenuse $= a^2 + b^2 = 130$
10	**(a)** 15	$18 \times 10 = 12 \times x$
11	**(d)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
12	**(b)** 17.5 km	3.5×5
13	**(b)** 20%	(SP−CP)/CP×100 = 300/1500×100
14	**(d)** 16, 63, 65	$2m = 16$, $m^2 - 1 = 63$, $m^2 + 1 = 65$
15	**(c)** 9	$135 \div 15$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$x^2 + 2 \times x \times 8 + 8^2$ (1) $= x^2 + 16x + 64$ (1).	1+1
17	3 m 60 cm $= 360$ cm (1); ratio $= 4 : 1$ (1).	1+1
18	$36/90 \times 100$ (1) $= 40\%$ (1).	1+1
19	Take $2m = 28$, so $m = 14$ (1); triplet $(28, 195, 197)$ (1).	1+1
20	$\frac{5}{9} = 0.56$ and $\frac{3}{5} = 0.6$ (1); so 3:5 is greater.	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$(2x)^2 + 2(2x)(3) + 3^2$ (2); $= 4x^2 + 12x + 9$ (1).	2+1
22	Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 4 : 6 : 5$ (2).	1+2
23	GST = 18% of 45000 = ₹ 8100 (2); bill = 45000 + 8100 = ₹ 53100 (1).	2+1
24	$2x^2 = (4\sqrt{2})^2 = 32$ (1); $x = 4$ cm (1); area $= \frac{1}{2} \times 4 \times 4 = 8$ cm² (1).	1+1+1
	OR: Length $= 14$ km (1); breadth $= 8$ km (1); perimeter $= 2(14+8) = 44$ km (1).	1+1+1
25	$53 = 50 + 3$, $47 = 50 - 3$ (1); product $= 2500 - 9$ (1); $= 2491$ (1).	1+1+1
	OR: Man-days $= 10 \times 18 = 180$ (1); days for 15 men $= 180 \div 15$ (1); 12 days (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Area $= (x+3)(x+5)$ (1); $= x^2 + 8x + 15$ (2); at $x = 4$: 63 cm² (2).	1+2+2
	OR: One part ₹ 150 (1); shares ₹ 150, ₹ 450, ₹ 600 (2); after the transfer ₹ 250, ₹ 450, ₹ 500 (1); new ratio $5 : 9 : 10$ (1).	1+2+1+1
27	A = P(1+R/100)³ = ₹ 10648 (2); CI = ₹ 2648 (1); SI = 8000×10×3/100 = ₹ 2400 (1); difference = ₹ 248 (1).	2+1+1+1
	OR: Half-diagonals 20 cm, 21 cm (1); side² $= 400 + 441 = 841$, side $= 29$ cm (2); perimeter $= 4 \times 29 = 116$ cm (1); area $= \frac{1}{2} \times 40 \times 42 = 840$ cm² (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** $x^2 + 7x + 10$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
2	**(a)** 10 days	Fewer or more workers: days vary inversely
3	**(c)** 20%	Discount = 500; 500/2500×100
4	**(a)** right-angled	$14^2 + 48^2 = 2500$ and $50^2 = 2500$
5	**(c)** 84	$14/60 \times 360$
6	**(d)** 9999	$(100 + 1)(100 - 1) = 100^2 - 1^2$
7	**(c)** 420 km	Speed $240 \div 4$
8	**(c)** 2100	GST = 100; bill = 2100
9	**(c)** $3\sqrt{2}$	$d^2 = s^2 + s^2 = 18$
10	**(c)** 900	One part $=$ ₹300
11	**(b)** 22	Add the two cross products
12	**(b)** ₹ 150	One part $= 150$
13	**(d)** 1050	Amount = 5000(1+10/100)² = 6050; CI = A − P
14	**(c)** 210	Other leg $= 21$ cm; area $= \frac{1}{2} \times 20 \times 21$
15	**(d)** 560	Speed $= 70$ km/h

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$6x^2 + 8x + 3x + 4$ (1) $= 6x^2 + 11x + 4$ (1).	1+1
17	Total work $= 9 \times 16 = 144$ worker-days (1); days $= 144 \div 12 = 12$ (1).	1+1
18	Loss = 15% of 24000 = ₹ 3600 (1); SP = ₹ 20400 (1).	1+1
19	$16^2 + 63^2 = 4225$ (1); $65^2 = 4225$ so the triangle is right-angled, the right angle is opposite the side 65 cm (1).	1+1
20	$xy = 6 \times 20 = 120$ (1); $y = 120 \div 8 = 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	A = P(1+R/100)² = 12500×108/100×108/100 = ₹ 14580 (2); CI = ₹ 2080 (1).	2+1
22	$d^2 = 29^2 - 21^2 = 841 - 441 = 400$ (2); $d = 20$ m (1).	2+1
23	$1\% = 3.6^\circ$ (1); food 108°, rent 90° (1); education 72°, others 90° (1).	1+1+1
24	$(x + 2)(x + 3) = x^2 + 5x + 6$ (1); $(x - 2)(x - 3) = x^2 - 5x + 6$ (1); sum $= 2x^2 + 12$ (1).	1+1+1
	OR: One part $= 6$ (1); present ages 18 and 30 (1); after 4 years 22 and 34 (1).	1+1+1
25	Change = 4000 (1); % = 4000/40000×100 (1) = 10% (1).	1+1+1
	OR: Diagonals bisect at right angles: half-diagonals 10 cm and 24 cm (1); side$^2 = 100 + 576 = 676$ (1); side $= 26$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Make B equal: A:B:C $= 8 : 12 : 15$ (2); total parts $= 35$, one part $=$ ₹150 (1); A = ₹ 1200, B = ₹ 1800, C = ₹ 2250 (2).	2+1+2
	OR: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 7$, $b = 3$: $100 - 16 = 84$ and $4ab = 84$ (2).	2+1+2
27	(i) speed $= 240 \div 4 = 60$ km/h (1.5); (ii) $60 \times 7 = 420$ km (1.5); (iii) $360 \div 60 = 6$ hours (2).	1½+1½+2
	OR: Discount = ₹ 3000 (1); discounted price = ₹ 27000 (1); GST = 18% of it = ₹ 4860 (1); payable = ₹ 31860 (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** $(a-b)^2 = a^2 - 2ab + b^2$	Check by expanding
2	**(b)** decreases	Inverse proportion
3	**(b)** 20%	Change/original × 100 = 50/250×100
4	**(c)** the square of the hypotenuse equals the sum of the squares of the other two sides	Statement of the theorem
5	**(a)** 5:8	Compare as fractions: $\frac{5}{8} \approx 0.625$, $\frac{7}{12} \approx 0.583$, $\frac{3}{5} \approx 0.600$, $\frac{4}{7} \approx 0.571$
6	**(d)** $x^2 + 8x + 16$	$(a+b)^2 = a^2 + 2ab + b^2$
7	**(c)** 5 : 1	Convert to the same unit
8	**(a)** 25%	35/140×100
9	**(a)** $2\sqrt{10}$	$h^2 = 2^2 + 6^2 = 40$
10	**(d)** 15:20:24	Make B equal: B $= \mathrm{LCM}(4,5) = 20$; A $= 15$, C $= 24$ gives $15 : 20 : 24$
11	**(b)** $x^2 - 10x + 25$	$(a-b)^2 = a^2 - 2ab + b^2$
12	**(a)** 6 : 8 : 7	Make the middle terms equal
13	**(b)** 10%	(SP−CP)/CP×100 = 240/2400×100
14	**(a)** 290	Square on hypotenuse $= a^2 + b^2 = 290$
15	**(a)** 12	$20 \times 9 = 15 \times x$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$48 \times 52 = (50-2)(50+2)$ (1) $= 2500 - 4 = 2496$ (1).	1+1
17	3.5 cm $\rightarrow$ 70 km (1); 100 km $\rightarrow$ 5 cm (1).	1+1
18	SI = ₹ 720 (1); R = SI×100/(P×T) = 6% p.a. (1).	1+1
19	$d^2 = 4^2 + 4^2 = 32$ (1); $d = 4\sqrt{2}$ cm (1).	1+1
20	$k = y/x = 35/7 = 5$ (1); $y = 5 \times 12 = 60$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Speed × time is constant: $30 \times 24 = 720$ (1); new speed $= 720 \div 18$ (1) $= 40$ km/h (1).	1+1+1
22	$(3x)^2 + 2(3x)(4) + 4^2$ (2); $= 9x^2 + 24x + 16$ (1).	2+1
23	Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 6 : 8 : 7$ (2).	1+2
24	GST = 12% of 36000 = ₹ 4320 (2); bill = 36000 + 4320 = ₹ 40320 (1).	2+1
	OR: $2x^2 = (6\sqrt{2})^2 = 72$ (1); $x = 6$ cm (1); area $= \frac{1}{2} \times 6 \times 6 = 18$ cm² (1).	1+1+1
25	Length $= 18$ km (1); breadth $= 15$ km (1); perimeter $= 2(18+15) = 66$ km (1).	1+1+1
	OR: $68 = 70 - 2$, $72 = 70 + 2$ (1); product $= 4900 - 4$ (1); $= 4896$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Difference in heights $= 20 - 12 = 8$ m (1); distance$^2 = 8^2 + 15^2 = 289$ (2); distance $= 17$ m (2).	1+2+2
	OR: Angle per mark $= 360° \div 60 = 6°$ (1); angles: Maths 108°, Science 90°, English 54°, Hindi 72°, SST 36° (3); Maths percentage $= 30\%$ (1).	1+3+1
27	Area $= (x+2)(x+6)$ (1); $= x^2 + 8x + 12$ (2); at $x = 5$: 77 cm² (2).	1+2+2
	OR: One part ₹ 200 (1); shares ₹ 400, ₹ 600, ₹ 800 (2); after the transfer ₹ 600, ₹ 600, ₹ 600 (1); new ratio $1 : 1 : 1$ (1).	1+2+1+1

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